The Lambda-Field: Drift Audit, Born Consistency, and the Measured Cost of Nonlinearity

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Abstract

> *This v3.1 audit applies to FRC 100.007's rho-level Lambda drift. Implementing the printed drift shows that the printed form is not trace-preserving for operational coherence functionals. The corrected anticommutator completion is trace-preserving by construction and preserves positivity in the pilot, but it moves Born populations linearly in alpha for both tested coherence choices. The result relocates outcome statistics to the microstate-distribution level used by FRC 100.006 and FRC 100.003. The same pilot also measures preparation dependence: a symmetric I/2 test is killed as degenerate, while a non-degenerate same-rho test gives Delta/alpha = 8.804 at small alpha. The*

> *Lambda-field architecture remains useful, but the rho-level nonlinear drift is an effective coherence-transport tool, not a Born-statistics carrier.*

1. Scope

This is an audit and correction note for the Lambda-field drift in FRC 100.007. It is not a full rewrite of the Lambda-field architecture.

What stands: the nesting principle, alpha → 0 recovery, no-fifth-force discipline, the Klein-Gordon relaxation layer, and the idea that Lambda can organize coherence geometry.

What changes: the rho-level drift cannot be used as printed, and the corrected rho-level drift cannot carry Born outcome statistics. It remains useful as an effective nonlinear coherence-transport model, but outcome statistics must live at the microstate-distribution level used in [[FRC-100-006]] and [[FRC-100-003]].

2. The architecture, briefly (as audited)

The earlier architecture paper (v8, DOI 10.5281/zenodo.17968952) defines the Lambda-field as $\Lambda(x) = \Lambda_0 \ln C(x)$, a real scalar field over the local coherence measure $C(x) \in (0,1]$, with a Klein-Gordon-style Lagrangian $\mathcal{L}_\Lambda = \tfrac{1}{2}(\partial_\mu\Lambda)(\partial^\mu\Lambda) - V(\Lambda)$, a Yukawa-like coupling to matter, and a screened-Poisson non-relativistic limit in which matter sources the field. That architecture — the field definition, the relaxation layer that gives a finite equilibration time, and the coupling structure — stands, subject to the audit below. What this version audits is the *rho-level nonlinear drift* that the v3 branch printed as the field's action on quantum states.

3. Form audit

FRC 100.007 v3 printed a rho-level drift of the form

$$\dot{\rho} = \mathcal{L}[\rho] + {\alpha\over \Lambda_0} \left(\nabla_\rho \Lambda - \mathrm{Tr}[\nabla_\rho\Lambda\,\rho] I\right).$$

For operational coherence functionals this is not trace-preserving. With $C_{super} = 2|\rho_{01}| + \epsilon$, the trace of the added term is not identically zero.

The corrected canonical completion used in the pilot is

$$\dot{\rho} = \mathcal{L}[\rho] + {\alpha\over \Lambda_0} \left({1\over 2}\{\nabla_\rho\Lambda, \rho\} - \mathrm{Tr}[\nabla_\rho\Lambda\,\rho]\,\rho\right).$$

This is the mixed-state analogue of a norm-preserving state-vector drift. It is trace-preserving identically. In the shipped pilot, the maximum trace error is 6.88e-15 and the minimum eigenvalue remains positive at 3.69e-4.

This is a canonical completion, not a uniqueness theorem. The important result below comes from the rho-dependence of the nonlinear drift, not from this closure alone.

4. Born audit

The pilot starts from a pure superposition with $|a_0|^2 = 0.64$, pure dephasing rate $\gamma = 0.2$, and the corrected rho-level drift. Two operational coherence choices are tested:

1. $C_{super} = 2|\rho_{01}| + \epsilon$, an inter-branch coherence channel.
2. $C_{lock} = (1 + |\langle\sigma_z\rangle|)/2$, a pointer-order channel.

Measured final populations:

Coherence choice	alpha	p0 final	Born deviation	early rate
C_super	0.00	0.640000	+0.000000	+0.00000/t
C_super	0.02	0.563245	-0.076755	-0.00253/t
C_super	0.05	0.519148	-0.120852	-0.00549/t
C_super	0.10	0.502611	-0.137389	-0.00874/t
C_lock	0.00	0.640000	+0.000000	+0.00000/t
C_lock	0.02	0.837754	+0.197754	+0.00652/t
C_lock	0.05	0.950928	+0.310928	+0.01413/t
C_lock	0.10	0.993315	+0.353315	+0.02248/t

The sign depends on the coherence channel. C_{super} pulls the populations toward the balanced superposition. C_{lock} pulls the populations toward a pole. Both violate Born statistics at the rho level.

The result is structural. A deterministic rho-level drift that climbs a coherence functional will generally move populations if that functional depends on populations. Therefore a rho-level nonlinear drift cannot be the carrier of outcome statistics unless the coherence functional is explicitly population-blind, in which case it cannot drive outcome selection.

This is why the outcome-statistics burden belongs to the microstate-distribution formulation: [[FRC-100-006]]'s continuity flow and [[FRC-100-003]]'s Gate-1 commitment.

5. Reconciliation flag

The v3 branch reportedly contained a figure in which $p_n \to |\alpha_n|^2$ under the drift. This audit contradicts that behavior for the printed rho-level drift and the corrected anticommutator completion.

The likely explanations are:

1. the plotted implementation used a population-blind gradient;
2. the non-trace-preserving printed form masked the drift after normalization;
3. alpha was too small for the population drift to be resolved;
4. the plotted process was already distribution-level rather than single-rho-level.

This must be reconciled before a future v3.2 claims Born recovery from the rho-level drift itself.

6. Nonlinearity and preparation dependence

The obvious mixture-linearity test is a trap. If $\rho = I/2$ is prepared as $\{|0\rangle, |1\rangle\}$ or $\{|+\rangle, |-\rangle\}$, the two averaged futures agree to machine precision even under the nonlinear drift (symmetric-test Delta ≈ 2.22e-16 for alpha = 0.01–0.10). This does not prove linearity. It is a degenerate-test kill: the decompositions are symmetric about the Bloch center, so the nonlinear changes cancel.

The non-degenerate test uses the same density matrix with Bloch $z = 0.4$, prepared two ways:

1. eigendecomposition: $0.7\,|0\rangle\langle 0| + 0.3\,|1\rangle\langle 1|$;
2. equal mixture of two pure states at Bloch $(\pm 0.917, 0, 0.4)$.

Under C_{super}, the two futures diverge:

alpha	Delta = trace norm difference	z_B final
0.01	0.088043	0.311957
0.02	0.156886	0.243114
0.05	0.284999	0.115001
0.10	0.366997	0.033003

The small-alpha protocol coefficient is

$${\Delta\over \alpha} \approx 8.804$$

at $t = 25$, $\gamma = 0.2$ in model units. Across the whole alpha range the least-squares slope is lower, 2.981, because the effect saturates. The small-alpha coefficient is the relevant number for bound protocols.

7. Alpha-bound protocols

The audit turns the nonlinearity concern into two concrete protocols:

1. **Born-precision bound:** population drift rates of order 0.09·alpha to 0.22·alpha per model-time unit at $\gamma = 0.2$ bound alpha from precision Born-frequency tests.
2. **Preparation-independence bound:** a test of decomposition independence at level delta bounds alpha ≤ delta / 8.804 in this model normalization.

No physical alpha number is asserted here. Physical translation requires the coupling normalization and experimental time map. That belongs in FRC 100.004 v2 or a dedicated constraints paper.

8. Ledger correction

The old closed ledger citation must be replaced by the open [[FRC-566-030]] frame. The closed statement

$$dS + k_*\,d\ln C = 0$$

is a boundary or locked-limit relation, not the general open-system law. For 100.007, Lambda drift should be read as a candidate transport model whose export channels and residuals must be measured, not assumed away.

9. What survives

The Lambda-field remains a useful corpus layer if kept in the right lane:

1. field relaxation and coherence geometry can be modeled;
2. alpha → 0 recovers ordinary quantum evolution;
3. the KG-style relaxation layer can define a finite equilibration time;
4. rho-level nonlinear drift can model effective coherence transport;
5. outcome statistics must be distribution-level, not single-rho-level;
6. nonlinear preparation dependence is now measurable and constrainable.

This strengthens 100.007 by pricing its risk instead of hiding it.

10. Kill conditions

This audit fails if:

1. the corrected drift fails trace preservation or positivity in the reference pilot;
2. the Born population drift disappears for both C_{super} and C_{lock};
3. the non-degenerate same-rho preparation test gives zero Delta beyond numerical tolerance;
4. a future v3.2 can reproduce Born recovery at the rho level with a population-sensitive coherence functional and no hidden distribution-level step.

The shipped pilot passes the audit checks and preserves the reconciliation flag.

Connections

- [[FRC-100-001]] — Original coherence definition and FRC operator
- [[FRC-100-003]] — Collapse as open-system phase-locking; Gate-1 commitment to distribution-level dynamics
- [[FRC-100-004]] — Experimental constraints; destination for physical alpha bounds
- [[FRC-100-006]] — Born rule from resonant equilibrium; the microstate-distribution flow that carries outcome statistics
- [[FRC-566-001]] — UCC as continuity equation for the Lambda current
- [[FRC-566-030]] — Open directional ledger replacing the closed-form citation

References

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3. H. Servat, *FRC 100.006 — Born Rule from Resonant Equilibrium.* DOI: 10.5281/zenodo.17438360.
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